

DOW UNIVERSITY OF HEALTH SCIENCES

BDS 1st Year

Session 2026

Foundation Module

Subject: Pharmacology

Learning Objectives

GENERAL PHARMACOLOGY	
1	INTRODUCTION TO PHARMACOLOGY
	ROUTES OF DRUG ADMINISTRATION
2	Describe various routes of drug administration (oral, parenteral, topical, and inhalational).
3	Compare advantages and disadvantages of different routes
4	Understand topical agents used on oral mucosa (e.g., gels, mouthwashes).
	PHARMACOKINETICS
	ABSORPTION OF DRUGS
5	Explain Mechanisms of Drug Transport and Give Examples
6	Define drug absorption and list factors affecting it.
7	Explain the role of pH, lipid solubility, and blood flow in drug absorption.
8	Describe first-pass metabolism
	BIOAVAILABILITY OF DRUGS
9	Define Bioavailability
10	Discuss Factors Affecting Bioavailability
	DISTRIBUTION OF DRUGS
11	Explain drug distribution and factors affecting it (plasma protein binding).
	BIOTRANSFORMATION (METABOLISM) OF DRUGS
12	Define plasma half-life and steady-state concentration.
13	Differentiate between first-order and zero-order kinetics.
14	Define drug metabolism and identify major metabolic organs.
15	Differentiate between Phase I and Phase II reactions
16	Explain enzyme induction and inhibition
	EXCRETION OF DRUGS
17	Describe routes of drug excretion
	FACTORS MODIFYING DRUG DOSE & ACTION / DOSAGE REGIMENS
18	List drug & patient-related factors affecting drug response (age, weight, disease).
	PHARMACODYNAMICS
	RECEPTORS, TYPES, ACTIONS & SIGNALING MECHANISMS
19	Define receptors and classify their types.
20	Explain drug-receptor interactions and signal transduction
	DRUG DOSE-RESPONSE RELATIONSHIP
21	Define dose-response relationship
22	Differentiate between graded and quantal responses
23	Explain therapeutic index and drug safety
24	Define agonists, antagonists, partial agonists.

25	Differentiate between competitive and non-competitive antagonism
26	Explain Different types of antagonism including pharmacological antagonism, physiological antagonism and chemical antagonism.